SQL COMMAND CATEGORIES

DDL • DML • TCL • DCL

Lecture Slides with Practical SQL Examples

Database Management System (DBMS)

Learning Objectives

Explain the purpose of DDL, DML, TCL, and DCL commands.

Identify common SQL commands in each category.

Write basic SQL statements using a Student table.

Understand transaction control and database permissions.

```
Example table used in this lecture: Student(StudentID, Name, Department, Email)
```

Four Categories of SQL Commands

Category	Full Form	Purpose	Common Commands
DDL	Data Definition Language	Define/change database structure	CREATE, ALTER, DROP, TRUNCATE
DML	Data Manipulation Language	Add or change table data	INSERT, UPDATE, DELETE
TCL	Transaction Control Language	Manage transactions	COMMIT, ROLLBACK, SAVEPOINT
DCL	Data Control Language	Manage permissions	GRANT, REVOKE

Note: SELECT is often taught as DQL (Data Query Language); some references group it under DML.

1. DDL — Data Definition Language

Used to create, modify, or remove database objects such as tables.

CREATE creates an object; ALTER changes its structure.

DROP removes an object; TRUNCATE removes all rows while keeping the table structure.

```
CREATE TABLE Student (
StudentID INT PRIMARY KEY,
    Name VARCHAR(50),
  Department VARCHAR(30)
          );

ALTER TABLE Student ADD Email VARCHAR(100);
```

DDL — More Examples

Command	Example	Effect
CREATE	CREATE TABLE Course (CourseID INT PRIMARY KEY);	Creates a table
ALTER	ALTER TABLE Student ADD Phone VARCHAR(20);	Adds a column
TRUNCATE	TRUNCATE TABLE Student;	Removes all rows; keeps structure
DROP	DROP TABLE Course;	Removes table definition and data

Caution: DROP and TRUNCATE can remove data. Confirm the target table before running them.

2. DML — Data Manipulation Language

Used to insert, update, and delete records in a table.

INSERT adds a row; UPDATE changes existing rows; DELETE removes rows.

Use WHERE with UPDATE and DELETE when you intend to affect selected records.

```
INSERT INTO Student (StudentID, Name, Department)
          VALUES (101, 'Amina', 'CSE');

UPDATE Student SET Department = 'IT' WHERE StudentID = 101;
        DELETE FROM Student WHERE StudentID = 101;
```

Reading Data: SELECT

SELECT retrieves data from a table. Many courses call this DQL (Data Query Language), although some classify it as part of DML.

```
SELECT * FROM Student;

SELECT Name, Department
     FROM Student
WHERE Department = 'CSE';
```

SELECT * returns all columns.

WHERE filters the rows returned.

For real tasks, select only the columns you need.

3. TCL — Transaction Control Language

Controls transactions: a group of operations treated as one unit.

COMMIT saves transaction changes.

ROLLBACK cancels uncommitted changes (subject to DBMS behavior).

SAVEPOINT creates a point to which you can roll back.

```
UPDATE Student SET Department = 'CSE' WHERE StudentID = 101;
                    SAVEPOINT before_change;
UPDATE Student SET Department = 'EEE' WHERE StudentID = 101;
                  ROLLBACK TO before_change;
                           COMMIT;
```

4. DCL — Data Control Language

Controls which database users can perform specific actions.

GRANT gives privileges; REVOKE removes privileges.

Exact syntax and privilege names can vary by database system.

```
-- Example syntax (availability depends on DBMS)
GRANT SELECT, INSERT ON Student TO analyst_user;

REVOKE INSERT ON Student FROM analyst_user;
```

Use least privilege: grant users only the access required for their job.

Putting the Categories Together

Step	Category	Example Task
1	DDL	CREATE TABLE Student (...)
2	DML	INSERT a student's details
3	DML / DQL	SELECT students in CSE
4	TCL	COMMIT or ROLLBACK a transaction
5	DCL	GRANT SELECT to an authorized user

Remember: DDL = structure | DML = data | TCL = transactions | DCL = permissions

Class Practice

1. Write SQL to create an Employee table with EmployeeID and EmployeeName.
2. Insert one employee record into the table.
3. Update the employee name for EmployeeID = 1.
4. Remove the record where EmployeeID = 1.
5. Which command saves a transaction? Which command grants access?

```
Answers: 1) CREATE TABLE  2) INSERT  3) UPDATE ... WHERE  4) DELETE ... WHERE
                                5) COMMIT; GRANT
```

KEY TAKEAWAYS

DDL	Defines database objects
DML	Manipulates table data
TCL	Manages transaction changes
DCL	Controls user privileges

Questions?